# Chapter ML:III

## III. Decision Trees

- Decision Trees Basics
- Impurity Functions
- Decision Tree Algorithms
- Decision Tree Pruning


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# Decision Tree Pruning

## Overfitting

### Definition 10 (Overfitting)

Let $D$ be a set of examples and let $H$ be a hypothesis space. The hypothesis $h \in H$ is considered to overfit $D$ if an $h' \in H$ with the following property exists:

$$\mathit{Err}(h, D) < \mathit{Err}(h', D) \quad \text{and} \quad \mathit{Err}^*(h) > \mathit{Err}^*(h'),$$

where $\mathit{Err}^*(h)$ denotes the true misclassification rate of $h$, while $\mathit{Err}(h, D)$ denotes the error of $h$ on the example set $D$.


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# Decision Tree Pruning

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Reasons for overfitting are often rooted in the example set $D$:

- $D$ is noisy and we "learn noise"
- $D$ is biased and hence not representative
- $D$ is too small and hence pretends unrealistic data properties


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# Decision Tree Pruning

## Overfitting (continued)

Let $D_{tr} \subset D$ be the training set. Then $Err^*(h)$ can be estimated with a test set $D_{ts} \subset D$ where $D_{ts} \cap D_{tr} = \emptyset$ [holdout estimation]. The hypothesis $h \in H$ is considered to overfit $D$ if an $h' \in H$ with the following property exists:

$$Err(h, D_{tr}) < Err(h', D_{tr}) \quad \text{and} \quad Err(h, D_{ts}) > Err(h', D_{ts})$$


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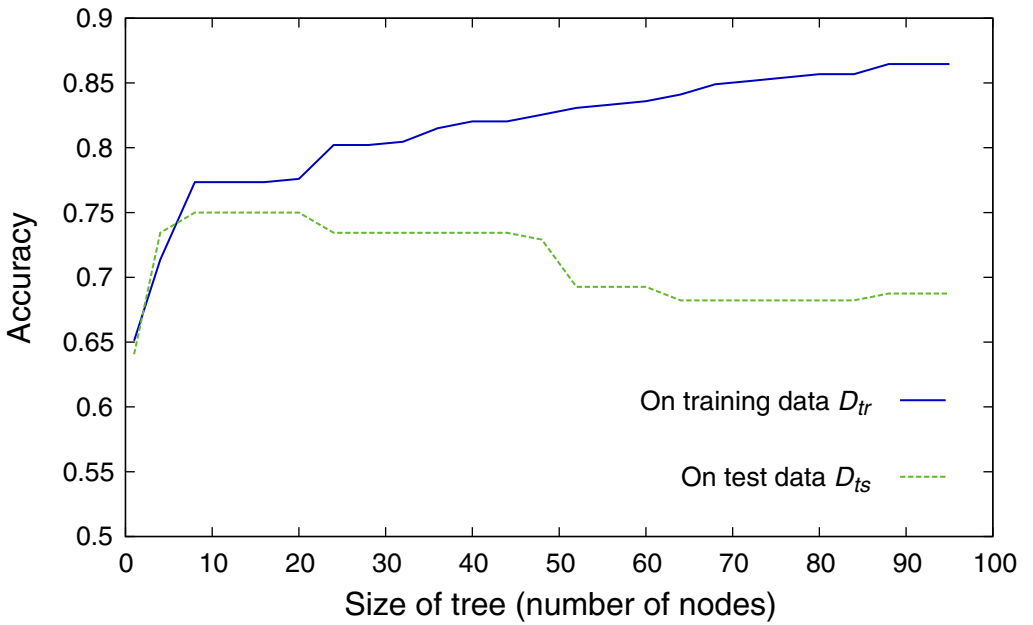


[Mitchell 1997]


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Remarks:

- Accuracy is the percentage of correctly classified examples.
- When does *Err*$(T, D_{tr})$ of a decision tree $T$ become zero?
- The training error *Err*$(T, D_{tr})$ of a decision tree $T$ is a monotonically decreasing function in the size of $T$. See the following Lemma.


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# Decision Tree Pruning

## Overfitting (continued)

### Lemma 10

Let $t$ be a node in a decision tree $T$. Then, for each induced splitting $D(t_1), \ldots, D(t_s)$ of a set of examples $D(t)$ holds:

$$\mathit{Err}_{\mathit{cost}}(t, D(t)) \geq \sum_{i \in \{1,\ldots,s\}} \mathit{Err}_{\mathit{cost}}(t_i, D(t_i))$$

The equality is given in the case that all nodes $t, t_1, \ldots, t_s$ represent the same class.


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# Decision Tree Pruning

## Overfitting (continued)

### Proof (sketch)

$$
\begin{aligned}
\mathit{Err}_{\mathit{cost}}(t, D(t)) &= \min_{c' \in C} \sum_{c \in C} p(c \mid t) \cdot p(t) \cdot \mathit{cost}(c' \mid c) \\
&= \sum_{c \in C} p(c, t) \cdot \mathit{cost}(\mathit{label}(t) \mid c) \\
&= \sum_{c \in C} (p(c, t_1) + \ldots + p(c, t_{k_s})) \cdot \mathit{cost}(\mathit{label}(t) \mid c) \\
&= \sum_{i \in \{1,\ldots,k_s\}} \sum_{c \in C} (p(c, t_i) \cdot \mathit{cost}(\mathit{label}(t) \mid c)
\end{aligned}
$$

$$
\mathit{Err}_{\mathit{cost}}(t, D(t)) - \textstyle\sum_{i \in \{1,\ldots,k_s\}} \mathit{Err}_{\mathit{cost}}(t_i, D(t_i)) =
$$

$$
\sum_{i \in \{1,\ldots,k_s\}} \left( \sum_{c \in C} p(c, t_i) \cdot \mathit{cost}(\mathit{label}(t) \mid c) - \min_{c' \in C} \sum_{c \in C} p(c, t_i) \cdot \mathit{cost}(c' \mid c) \right)
$$

Observe that the summands on the right equation side are greater than or equal to zero.


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Remarks:

- The lemma does also hold if the misclassification rate is used to evaluate effectiveness.
- The algorithm template for the construction of decision trees, *DT-construct*, prefers larger trees, entailing a more fine-grained partitioning of $D$. A consequence of this behavior is a tendency to overfitting.


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# Decision Tree Pruning

## Overfitting (continued)

Approaches to counter overfitting:

(a) Stopping of the decision tree construction process during training.

(b) Pruning of a decision tree after training:

- Partitioning of $D$ into three sets for training, validation, and test:
  - reduced error pruning
  - minimal cost complexity pruning
  - rule post pruning
- statistical tests such as $\chi^2$ to assess generalization capability
- heuristic pruning


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# Decision Tree Pruning

## (a) Stopping

Possible criteria for stopping [splitting criteria]:

1. Size of $D(t)$.
   $D(t)$ will not be partitioned further if the number of examples, $|D(t)|$, is below a certain threshold.

2. Purity of $D(t)$.
   $D(t)$ will not be partitioned further if all induced splittings yield no significant impurity reduction $\Delta\iota$.

Problems:

ad 1) A threshold that is too small results in oversized decision trees.

ad 1) A threshold that is too large omits useful splittings.

ad 2) $\Delta\iota$ cannot be extrapolated with regard to the tree height.


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# Decision Tree Pruning

## (b) Pruning

The pruning principle:

1. Construct a sufficiently large decision tree $T_{\text{max}}$.
2. Prune $T_{\text{max}}$, starting from the leaf nodes upwards to the tree root.

Each leaf node $t$ of $T_{\text{max}}$ fulfills one or more of the following conditions:

- $D(t)$ is sufficiently small. Typically, $|D(t)| \leq 5$.
- $D(t)$ is comprised of examples of only one class.
- $D(t)$ is comprised of examples with identical feature vectors.


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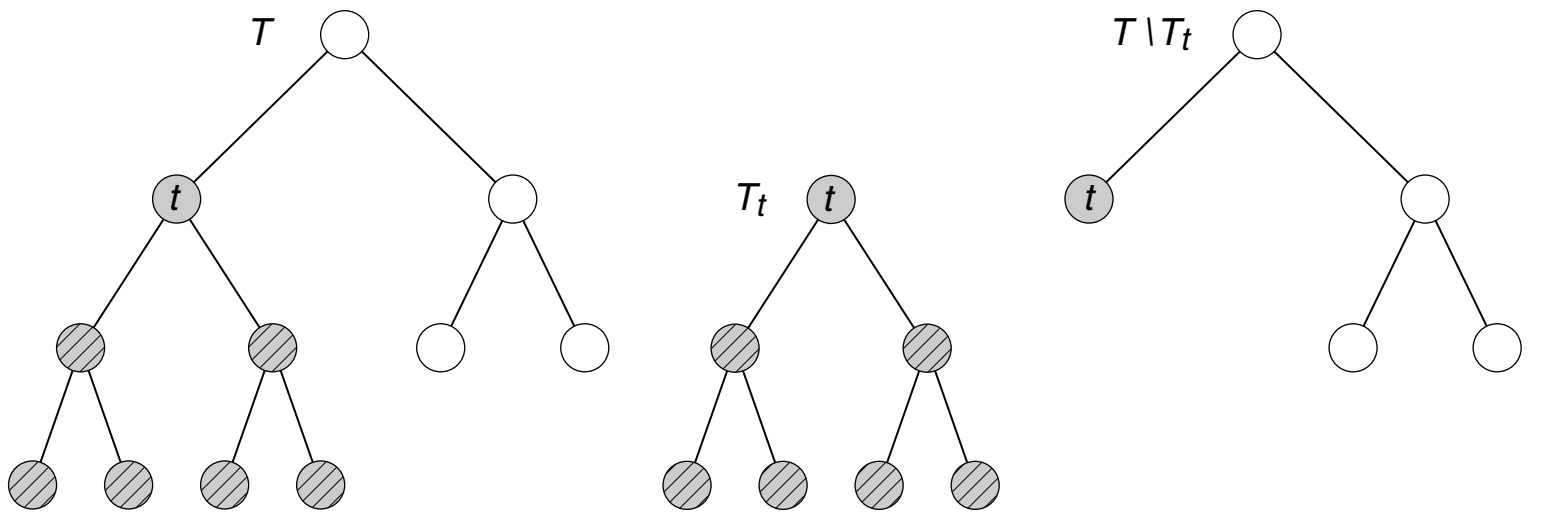
# Decision Tree Pruning

## (b) Pruning (continued)

**Definition 11 (Decision Tree Pruning)**

Given a decision tree $T$ and an inner (non-root, non-leaf) node $t$. Then pruning of $T$ with regard to $t$ is the deletion of all successor nodes of $t$ in $T$. The pruned tree is denoted as $T \setminus T_t$. The node $t$ becomes a leaf node in $T \setminus T_t$.

Illustration:





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# Decision Tree Pruning

## (b) Pruning (continued)

**Definition 12 (Pruning-Induced Ordering)**

Let $T'$ and $T$ be two decision trees. Then $T' \preceq T$ denotes the fact that $T'$ is the result of a (possibly repeated) pruning applied to $T$. The relation $\preceq$ forms a partial ordering on the set of all trees.


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Problems when assessing pruning candidates:

- Pruned decision trees may not stand in the $\preceq$-relation.
- Locally optimum pruning decisions may not result in the best candidates.
- Its monotonicity disqualifies $\mathit{Err}(T, D_{tr})$ as an estimator for $\mathit{Err}^*(T)$. [Lemma]


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Control pruning with validation set $D_{vd}$, where $D_{vd} \cap D_{tr} = \emptyset$, $D_{vd} \cap D_{ts} = \emptyset$:

1. $D_{tr} \subset D$ for decision tree construction.
2. $D_{vd} \subset D$ for overfitting analysis *during* pruning.
3. $D_{ts} \subset D$ for decision tree evaluation *after* pruning.


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# Decision Tree Pruning

## (b) Pruning: Reduced Error Pruning

Steps of reduced error pruning :

1. $T = T_{\text{max}}$
2. Choose an inner node $t$ in $T$.
3. Perform a tentative pruning of $T$ with regard to $t$ : $T' = T \setminus T_t$.
   Based on $D(t)$ assign class to $t$. [*DT-construct*]
4. If *Err*$(T', D_{vd}) \leq$ *Err*$(T, D_{vd})$ then accept pruning: $T = T'$.
5. Continue with Step 2 until all inner nodes of $T$ are tested.


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# Decision Tree Pruning

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   Based on $D(t)$ assign class to $t$. [*DT-construct*]
4. If *Err*$(T', D_{vd}) \leq$ *Err*$(T, D_{vd})$ then accept pruning: $T = T'$.
5. Continue with Step 2 until all inner nodes of $T$ are tested.

Problem:

If $D$ is small, its partitioning into three sets for training, validation, and test will discard valuable information for decision tree construction.

Improvement: rule post pruning


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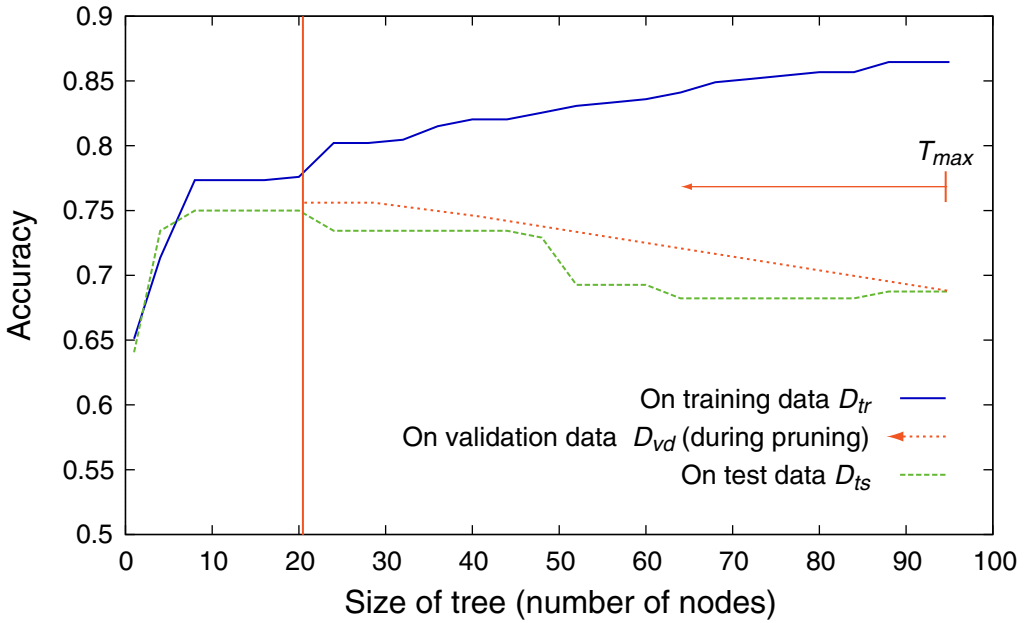
# Decision Tree Pruning

## (b) Pruning: Reduced Error Pruning (continued)

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# Decision Tree Pruning

## Extensions

- consideration of the misclassification cost introduced by a splitting
- “surrogate splittings” for insufficiently covered feature domains
- splittings based on (linear) combinations of features
- regression trees


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